Advanced SQL Assignment

Q1. What is a Common Table Expression (CTE)?

A Common Table Expression (CTE) is a temporary result set that is defined using the `WITH` keyword and is used within a query. It helps in simplifying complex SQL queries.

Explanation:
CTEs make queries easier to understand because instead of writing long nested queries, we can divide them into smaller parts. It also improves readability and makes debugging easier.

Syntax:

```
WITH cte_name AS (
   SELECT column_name FROM table_name
)
SELECT * FROM cte_name;
```

Q2. Why are some views updatable while others are read-only?

Views are virtual tables created using SQL queries.

- **Updatable views** allow us to insert, update, or delete data.
- **Read-only views** do not allow modification.

Explanation:
A view becomes read-only when it uses complex operations like GROUP BY, JOIN, DISTINCT, or aggregate functions. Simple views created from a single table are usually updatable.

Example:

```
CREATE VIEW simple_view AS
SELECT ProductID, ProductName FROM Products;

CREATE VIEW complex_view AS
SELECT Category, COUNT(*) FROM Products GROUP BY Category;
```

Q3. Advantages of Stored Procedures

Stored procedures are pre-written SQL queries stored in the database.

Advantages:

- They can be reused multiple times
- They improve performance
- They reduce repetition of code
- They provide better security
- They are easy to maintain

Q4. What is the purpose of Triggers?

A trigger is a special type of stored program that runs automatically when an event occurs on a table.

Explanation:
Triggers are used to maintain data consistency. For example, when a record is deleted, a trigger can automatically store it in another table.

Q5. Explain Data Modelling and Normalization

Data Modelling:
It is the process of designing the structure of a database.

Normalization:
It is used to remove duplicate data and organize data properly.

Importance:

- Reduces redundancy
- Improves data accuracy
- Makes database efficient

DELIMITER ;
